

Security in Informatics and Communications (2025/2026)

Practical Class (#7):

Public Key Infrastructure (PKI)

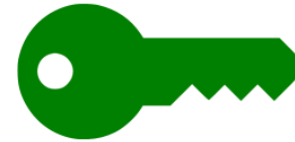
Asymmetric Cryptography

Private Key



- Sign data
- Decipher ciphered data

Public Key



- Verify signed data
- Cipher data

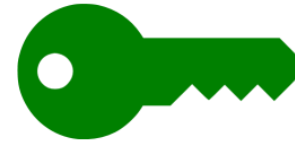
Asymmetric Cryptography

Private Key



- Sign data
- Decipher ciphered data

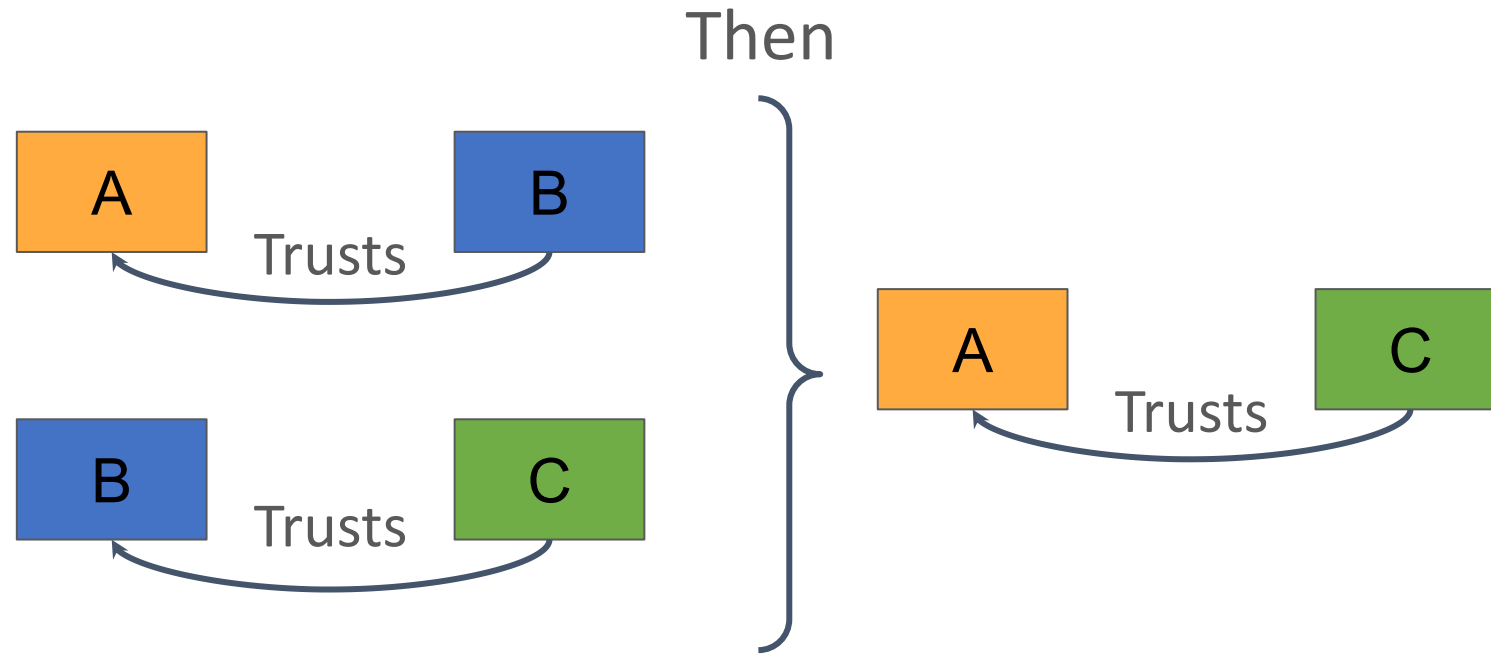
Public Key



- Verify signed data
- Cipher data

However, how can I trust a Public Key?

Trust



trust is transitive

How Trust works

Trust is transitive:

$$\text{Trust}(A, B) \wedge \text{Trust}(B, C) \Rightarrow \text{Trust}(A, C)$$

I *trust* my Bank **and** my Bank *trusts* the Bank Clerk , **therefore** I *trust* the Bank Clerk

But does not work the other way around:

$$\text{Trust}(A, C) \not\Rightarrow \text{Trust}(A, B) \wedge \text{Trust}(B, C)$$

(it may be true in some cases, but that's just chance!)

I *trust* my Bank ,  **therefore** I *trust* some Criminal **and** the Criminal *trusts* my Bank

Trusted Roots



ORDEM DOS
ADVOGADOS



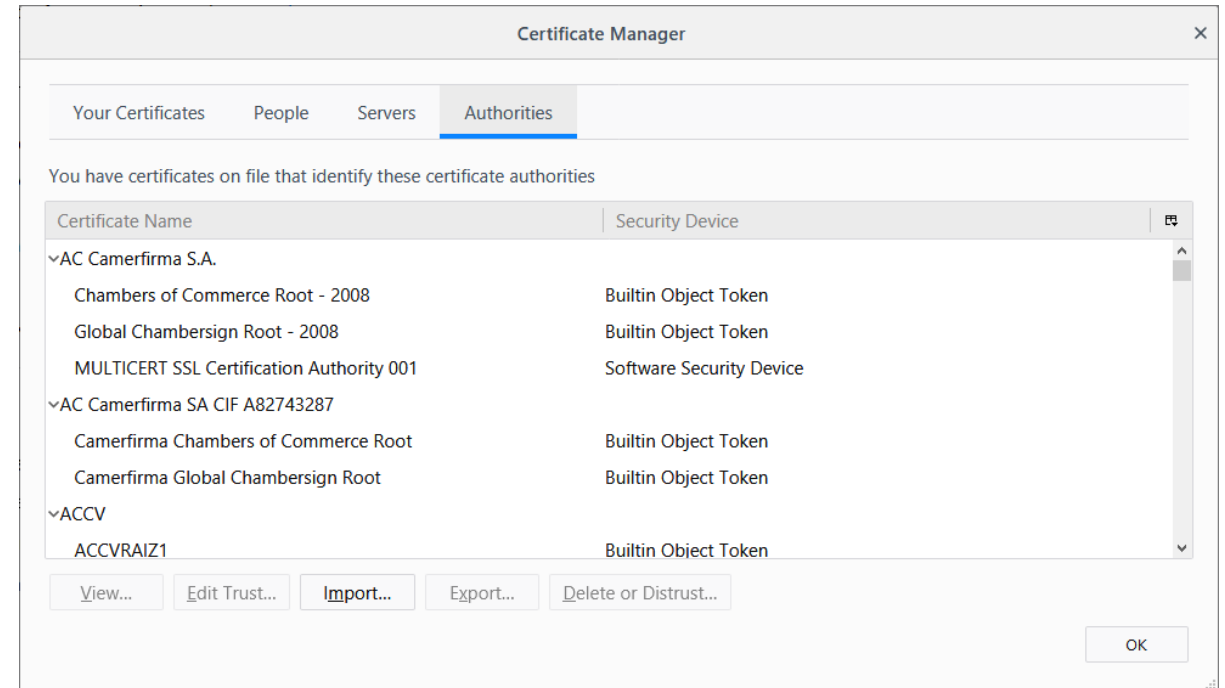
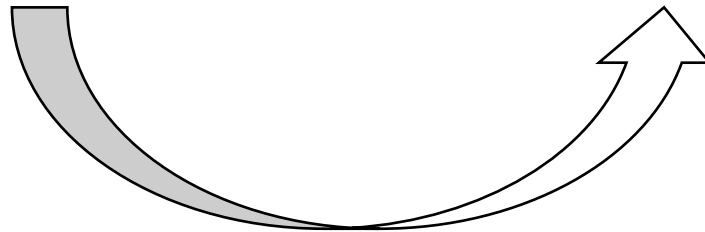
IMT INSTITUTO DA
MOBILIDADE E DOS
TRANSPORTES, I.P.



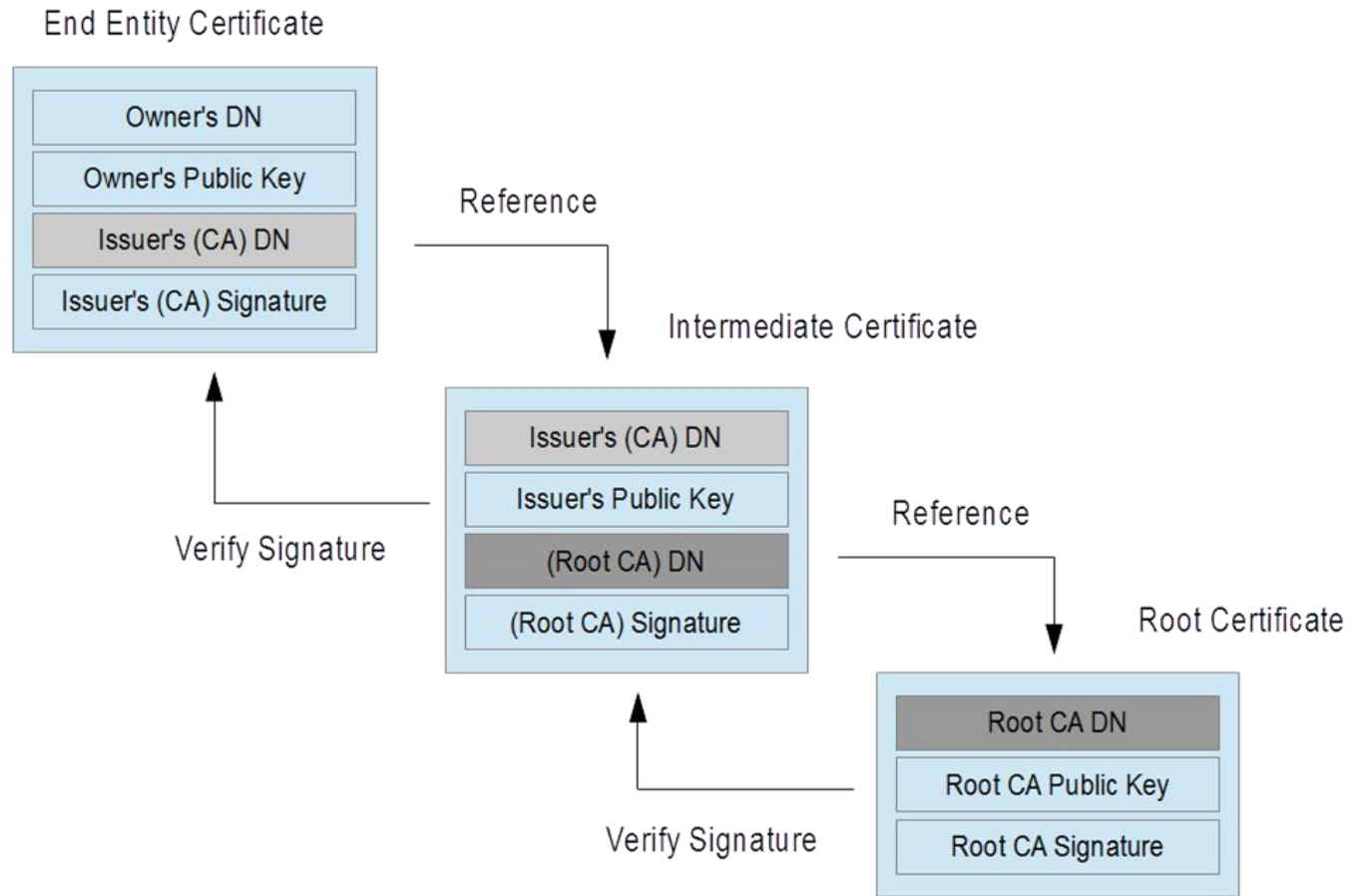
instituto dos
registos
e do notariado



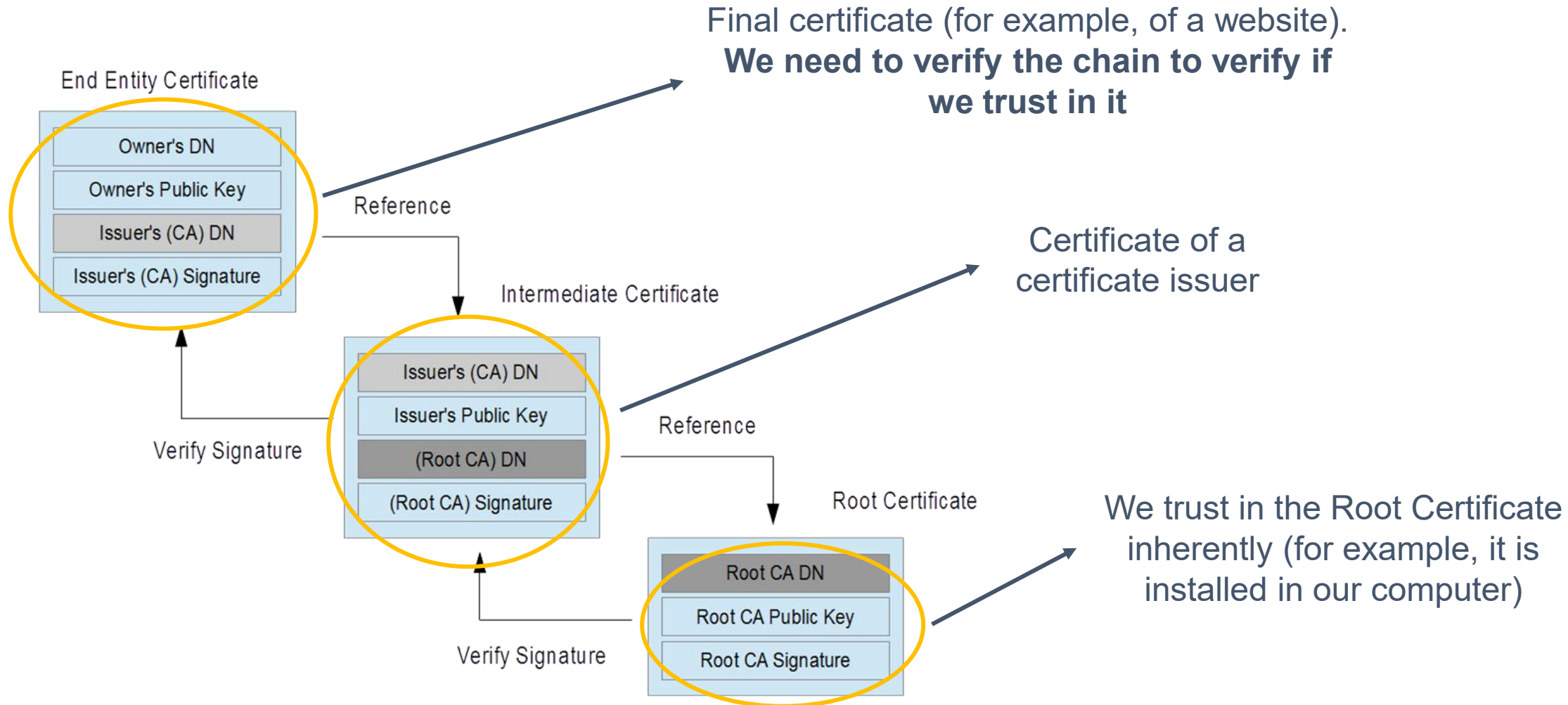
ORDEM
DOS
ENGENHEIROS



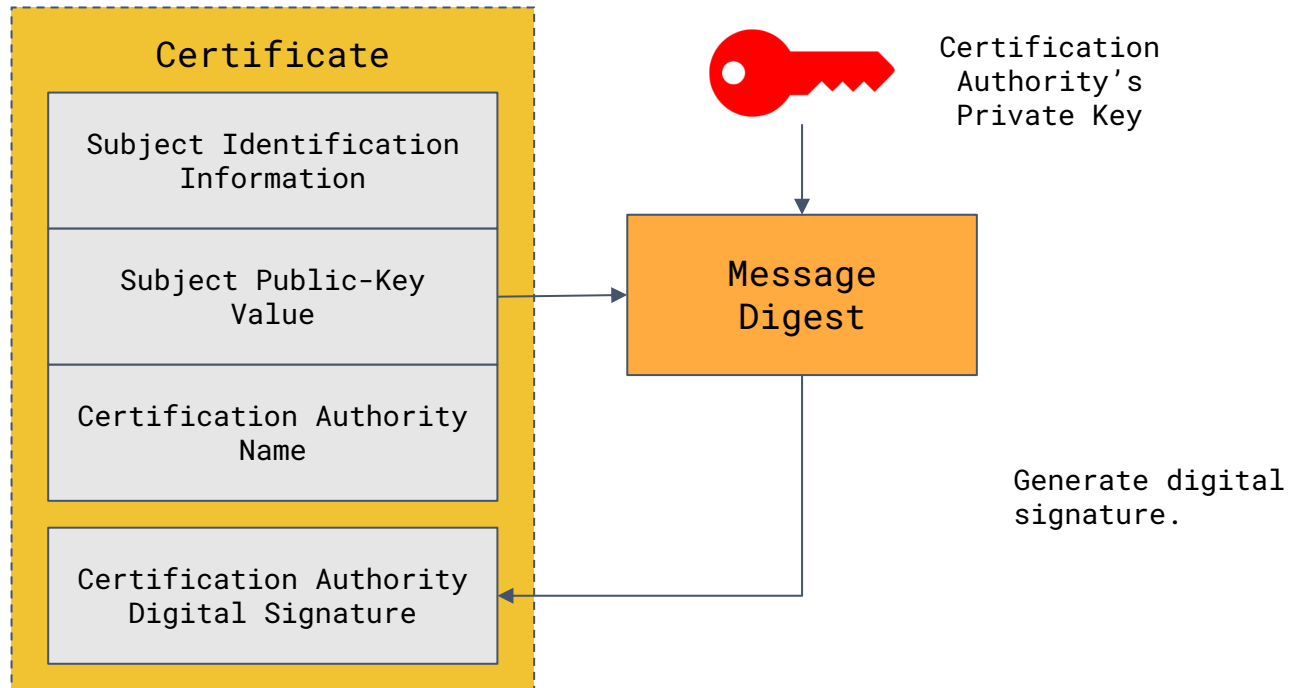
Chain of Trust



Chain of Trust



Trusting a Public Key



A certificate is just a format, to convey the trust of an authority on a specific key.

Public Key Certificates (X.509)

Certificate:

Data:

Version: 3 (0x2)

Serial Number:

10:e6:fc:62:b7:41:8a:d5:00:5e:45:b6

Signature Algorithm: sha256WithRSAEncryption

Issuer: C=BE, O=GlobalSign nv-sa, CN=GlobalSign Organization Validation CA - SHA256 - G2

Validity

Not Before: Nov 21 08:00:00 2016 GMT

Not After : Nov 22 07:59:59 2017 GMT

Subject: C=US, ST=California, L=San Francisco, O=Wikimedia Foundation, Inc., CN=*.wikipedia.org

Subject Public Key Info:

Public Key Algorithm: id-ecPublicKey

Public-Key: (256 bit)

pub:

04:c9:22:69:31:8a:d6:6c:ea:da:c3:7f:2c:ac:a5:

af:c0:02:ea:81:cb:65:b9:fd:0c:6d:46:5b:c9:1e:

ed:b2:ac:2a:1b:4a:ec:80:7b:e7:1a:51:e0:df:f7:

c7:4a:20:7b:91:4b:20:07:21:ce:cf:68:65:8c:c6:

9d:3b:ef:d5:c1

ASN1 OID: prime256v1

NIST CURVE: P-256

Public Key Certificates (X.509)

X509v3 extensions:

X509v3 Key Usage: critical

Digital Signature, Key Agreement

Authority Information Access:

CA Issuers - URI:<http://secure.globalsign.com/cacert/gsorganizationvalsha2g2r1.crt>

OCSP - URI:<http://ocsp2.globalsign.com/gsorganizationvalsha2g2>

X509v3 Certificate Policies:

Policy: 1.3.6.1.4.1.4146.1.20

CPS: <https://www.globalsign.com/repository/>

Policy: 2.23.140.1.2.2

X509v3 Basic Constraints:

CA:FALSE

X509v3 CRL Distribution Points:

Full Name:

URI:<http://crl.globalsign.com/gs/gsorganizationvalsha2g2.crl>

X509v3 Subject Alternative Name:

DNS:*.wikipedia.org, DNS:*.m.mediawiki.org, www, DNS:wikipedia.org

X509v3 Extended Key Usage:

TLS Web Server Authentication, TLS Web Client Authentication

X509v3 Subject Key Identifier:

28:2A:26:2A:57:8B:3B:CE:B4:D6:AB:54:EF:D7:38:21:2C:49:5C:36

X509v3 Authority Key Identifier:

keyid:96:DE:61:F1:BD:1C:16:29:53:1C:C0:CC:7D:3B:83:00:40:E6:1A:7C

Public Key Certificates (X.509)

Signature Algorithm: sha256WithRSAEncryption

8b:c3:ed:d1:9d:39:6f:af:40:72:bd:1e:18:5e:30:54:23:35:

...

Validation:



Public Key Certificates (X.509)

X509v3 extensions:

X509v3 Key Usage: critical

Digital Signature, Key Agreement

Authority Information Access:

CA Issuers - URI:<http://secure.globalsign.com/cacert/gsorganizationvalsha2g2r1.crt>

OCSP - URI:<http://ocsp2.globalsign.com/gsorganizationvalsha2g2>

X509v3 Certificate Policies:

Policy: 1.3.6.1.4.1.4146.1.20

CPS: <https://www.globalsign.com/repository/>

Policy: 2.23.140.1.2.2

X509v3 Basic Constraints:

CA:FALSE

X509v3 CRL Distribution Points:

Full Name:

URI:<http://crl.globalsign.com/gs/gsorganizationvalsha2g2.crl>

X509v3 Subject Alternative Name:

DNS:*.wikipedia.org, DNS:*.m.mediawiki.org, ... DNS:wikipedia.org

X509v3 Extended Key Usage:

TLS Web Server Authentication, TLS Web Client Authentication

X509v3 Subject Key Identifier:

28:2A:26:2A:57:8B:3B:CE:B4:D6:AB:54:EF:D7:38:21:2C:49:5C:36

X509v3 Authority Key Identifier:

keyid:96:DE:61:F1:BD:1C:16:29:53:1C:C0:CC:7D:3B:83:00:40:E6:1A:7C

Any questions?